Course code DATA72000

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Feature Representation, Cluster Structure and Explanation Tradeoffs in Customer Segmentation

A report submitted to The University of Manchester for the degree of Master of Science in the Faculty of Humanities

2026

School of Social Sciences

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Final word count including footnotes and endnotes 9193

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Abstract

Customer segmentation groups people using measured attributes, so the choice of those attributes may change both the groups and their explanations. This dissertation compares purchase summaries with a richer set of behavioural features for the same 11,719 RetailRocket purchasers. It applies K Means and CLASSIX under a shared evaluation design, then examines the resulting assignments, their repeatability, and the information needed to describe them. CLASSIX provides a record of how its groups were formed; ExKMC provides decision rules that approximate K Means assignments.

The two representations produce substantially different customer groups, even though repeated fitting within each representation generally gives similar assignments. Adding behavioural features also changes how accurately short decision rules reproduce the groups and how detailed the CLASSIX construction record becomes. Only one selected cluster meets the study's operational checks for size, data coverage and distinctive features. Sensitivity analyses qualify these findings under changes to features, numerical inputs and session definitions.

Representation is therefore part of the clustering model rather than a preliminary choice without consequences. In this dataset, richer information changes the grouping and its explanation, but does not consistently improve geometric quality or yield groups ready for operational use. Human understanding, campaign effectiveness and realised business value remain unmeasured.

Declaration

This report is my original work unless referenced clearly to the contrary. No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Intellectual Property Statement

AI Statement

Artificial intelligence tools were used as assistants for code drafting, data processing, numerical checks, source discovery and English language polishing. The author determined the research design, verified all sources and results, reviewed every substantive claim and accepts responsibility for the report. Details relevant to reproducibility are recorded in the accompanying materials.

Chapter 1

Introduction

1.1 Research background

Customer segmentation groups people according to recorded characteristics. Those characteristics might describe purchases, browsing or other interactions; being a customer does not necessarily mean making a purchase. This study deliberately examines only people with a recorded purchase in the RetailRocket dataset. Its conclusions concern that population.

A clustering algorithm receives a feature vector: a numerical description of each person. The *representation* is the choice and construction of those numerical features. In this study, a compact representation describes purchase timing and frequency, while a rich representation adds browsing activity, conversion, timing and category measures. These choices change the *feature geometry*: the distances and neighbourhood relationships between observations. Two people who appear similar in purchase frequency may differ substantially in browsing or category behaviour. An algorithm that groups nearby vectors can therefore assign the same people differently when those dimensions are added.

Comparing only the final clustering score can obscure this dependence. If both the population and the features change, the source of a different partition is unclear. A *partition* here is the assignment of the analysed observations to groups, with a separate noise state where the method permits unassigned observations. Clusterability research asks whether a dataset contains structure suitable for clustering, but gives no single definition that applies to every type of structure (Ackerman and Ben-David, 2009; Adolfsson, Ackerman and Brownstein, 2019). The relevant question is how the represented information changes the observed grouping under a specified design.

Explaining a grouping raises a related distinction. CLASSIX forms small local groups and then merges them; its *provenance* is the recorded route from an observation through those groups to its final cluster (Chen and Güttel, 2024). K Means instead assigns observations to nearby centres. ExKMC builds a decision tree after a K Means reference has been fitted, using single-feature thresholds to approximate its assignments (Frost, Moshkovitz and Rashtchian, 2020). The first explanation describes a construction process; the second describes how well a set of rules reproduces a reference. Neither establishes that the groups are useful to a business or understood by a person.

1.2 Research questions and contribution

The comparison holds the 11,719 purchasers, event snapshot, common purchase variables, candidate grids and evaluation rules fixed. The rich representation retains the compact representation's three features and adds nine behavioural features. Three questions organise the existing empirical programme:

1. The effect of changing from compact to rich features on customer assignments, geometric quality and repeatability for K Means and CLASSIX.
2. The effect of representation on the CLASSIX construction trace and the accuracy of ExKMC rules at a given rule budget, together with the extent to which selected clusters meet the stated operational checks.
3. The sensitivity of these findings to feature removal, reduced dimension, small numerical changes and alternative session definitions.

The contribution is a paired representation comparison that follows the same customers through clustering, explanation and operational assessment. It separates repeatability within one feature space from agreement between spaces, and separates intrinsic provenance from post hoc rule approximation. The robustness analyses test specific dependencies of the observed findings. A complementary mathematical contribution explains why CLASSIX can skip some distance calculations and states local sufficient conditions for an unchanged distance-merging fit. These conditions do not prove stability for all CLASSIX configurations.

This design does not identify objectively correct customer segments. It tests how observed assignments and their explanations depend on representation in one dataset. Neither additional features nor a higher internal score is assumed to imply better segmentation or realised business value.

1.3 Dissertation structure

Chapter 2 positions the study within research on representation, clustering validation and explanation. Chapter 3 defines the features, algorithms and evaluation measures before explaining the mathematical claims. Chapter 4 first reports a benchmark on nine labelled datasets. Alongside K Means and CLASSIX, this includes DBSCAN, which groups connected dense regions and can leave sparse observations unassigned, and Ward's method, which successively merges groups to limit the increase in within-group squared dispersion (Ester et al., 1996; Ward, 1963). Benchmark labels are used for evaluation, not for selecting parameters.

The main customer results then answer the three research questions through partition comparisons, repeated subsampling, explanation and operational checks, followed by sensitivity analyses. In these customer comparisons, one fitted partition is compared with another; neither is treated as ground truth. Chapter 5 draws a conclusion limited to that evidence.

Chapter 2

Literature Review

2.1 Representation in customer segmentation

An event log is a collection of timestamped actions, such as a visitor viewing an item, adding it to a cart or purchasing it. Converting these records into a feature table determines which similarities a clustering algorithm can use. Representation would be neutral only if those choices left the relevant grouping unchanged; that cannot be assumed merely because the same algorithm is used. A customer can be described by a few purchase summaries, the full sequence of recorded actions, or a rich set of temporal, conversion and category signals. Each choice changes the distances observed by the algorithm. Clickstream research describes raw action sequences as large, variable and noisy, then frames useful segments as subsets recovered through constraints on derived attributes (Dextras-Romagnino and Munzner, 2019). Work on object centred event logs likewise converts traces into vector profiles before clustering and pattern analysis (Faria Junior et al., 2023). Customer behaviour studies construct features for segmentation and later decision tasks, but their application specific choices do not establish a universal representation (Abdi and Abolmakarem, 2019). Representation is part of the empirical model rather than a preliminary clerical step.

2.2 Cluster structure and validation

A partition groups observations; validation asks what evidence supports that grouping. Internal validation uses the geometry of the features, whereas external validation compares assignments with separately supplied labels. Repeated fitting asks whether assignments recur under specified changes to the input. None of these questions can substitute for the others. Clustering does not reveal a representation independent truth. Clusterability theory supplies several incompatible formal notions of when data possess recoverable structure (Ackerman and Ben-David, 2009). Empirical comparisons also show that clusterability measures depend on distributional conditions, dimension and the type of structure sought (Adolfsson, Ackerman and Brownstein, 2019). A high internal score supports a partition within a defined feature space, not an intrinsic segmentation outside that space. Dimensional reduction retains this dependence. It replaces named variables with a new geometry and can obscure the link between a cluster and the original attributes (Bertsimas, Orfanoudaki and Wiberg, 2021). A paired comparison on the same customers is more informative than results drawn from unrelated populations.

2.3 Explainable clustering

Explainable clustering contains three distinct design positions. Intrinsic methods expose the computations that create a partition. CLASSIX sorts observations along a principal direction, aggregates nearby points and merges the resulting groups (Chen and Güttel, 2024). Its public group assignments, starting points and final labels support a trace from an observation to a cluster. Explainability first methods instead make a simple description part of the clustering objective. ICOT constructs tree partitions through optimisation, while XClusters balances distortion against the size and accuracy of a tree during cluster formation (Bertsimas, Orfanoudaki and Wiberg, 2021; Hwang and Whang, 2023). Post hoc methods begin with a reference partition and approximate it with rules or a tree. These positions explain different objects. Provenance from an intrinsic algorithm answers how its own grouping arose. A post hoc tree answers how closely axis aligned rules can reproduce another partition. Neither answer can be converted into the other without changing the target of explanation.

Table 2.1 Comparison of explanation designs in the reviewed literature

Method	Target	Explanation stage	Form	Complexity control	Status
CLASSIX	Own partition	Intrinsic	Groups and starting points	Radius and minimum size	Journal
IMM	K Means objective	Tree construction	Threshold tree	One leaf per cluster	Conference
ExKMC	K Means labels	Post hoc	Expanded tree	Leaf budget	Preprint
CREAM	Data or model outputs	During induction	Cubes and tree rules	Maximum depth and error threshold	Conference

Method	Target	Explanation stage	Form	Complexity control	Status
ICOT	Own partition	Joint	Optimised tree	Validation objective and tree	Journal
XClusters	Joint objective	Joint	Decision tree	Distortion and node tradeoff	Conference

2.4 Explanation complexity and fidelity

Decision trees make the tradeoff between concise expression and partition quality explicit. IMM uses threshold cuts to construct a small tree and analyses the additional K Means or K Medians cost caused by that restriction (Moshkovitz et al., 2020). Theoretical work shows that such simplicity can impose unavoidable objective loss and that the magnitude depends on the clustering problem, cluster count and dimension (Laber and Murtinho, 2021). Other work asks how a fixed clustering can be made explainable after removing a limited number of outlying observations, which further demonstrates that an apparently simple tree may obtain its clarity by changing the population being explained (Bandyapadhyay et al., 2022). CREAM constructs rectangular regions through tree induction, using clustering within the construction and permitting model-output approximation in supervised settings. Its maximum depth and predictive-error threshold govern further splitting (Sabbatini and Calegari, 2023). These methods make complexity measurable through leaves, depth, conditions and features, but those counts do not by themselves establish that a person will understand or use the explanation.

The guarantees concern different quantities. IMM bounds the extra distance-based clustering cost imposed by a tree; such a bound does not say how many people agree with its labels. CLASSIX exposes intermediate groups from its own computation; that record does not impose a short decision rule. ICOT and XClusters make tree structure part of cluster formation, so changing the explanation can also change the partition being explained. CREAM limits the refinement of its rectangular regions through depth and error controls. These distinctions explain the present choice to retain separate provenance and tree-approximation evaluations rather than rank the methods on one explanation scale.

ExKMC expands a tree beyond one leaf for each K Means cluster, allowing the leaf budget to test whether additional rules improve agreement with a fixed reference. The source used here is the authors' arXiv preprint; its cost analysis motivates the leaf-budget comparison, but does not guarantee held-out label agreement (Frost, Moshkovitz and Rashtchian, 2020). Fidelity here means exact predictive agreement between the tree and held out K Means assignments. It is neither an internal clustering metric nor evidence that the K Means partition is substantively correct. More general XAI research finds that fidelity measures can disagree even when a transparent model supplies a known reference, so the target and calculation must be stated directly (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

2.5 Research gap and positioning

Operational usefulness introduces another boundary. A tree or a set of threshold conditions may be machine queryable, while an intrinsic trace may identify prototypes and merge relations. Queryability can be observed through the number of conditions, the use of named features, customer coverage and agreement with a reference partition. It does not establish improved revenue, better interventions or human comprehension. Segmentifier illustrates why recorded constraints and provenance can support the refinement and export of behavioural segments, but its design study does not validate the commercial effect of a clustering model (Dextras-Romagnino and Munzner, 2019). Similarly, theoretical guarantees on objective cost do not measure whether a segment is large enough, distinct enough or sufficiently covered by reliable attributes to be acted upon.

The gap addressed here is a controlled link between representation choice, partition change and distinct explanation properties. The reviewed methods supply useful constructions and objectives, but do not by themselves answer whether adding behavioural features to the same purchaser cohort improves all these properties together. The present study compares compact transaction and rich behavioural representations for the same RetailRocket purchasers under matched clustering and evaluation conditions, and relates differences in cluster structure to CLASSIX provenance, ExKMC fidelity and auditable addressability.

Chapter 3

Data and Methodology

This chapter explains how the RetailRocket event data are converted into two customer representations and evaluated under matched analytical conditions. It also distinguishes the mathematical properties of CLASSIX from the empirical robustness checks conducted in this study.

3.1 Research design

The research design compares two descriptions of the same purchasers. The compact representation contains three variables that summarise purchase timing and frequency. The rich representation contains twelve variables. It retains the same three purchase variables and adds nine measures of activity, conversion, temporal engagement and category behaviour. Both representations are constructed from the same event history for the same 11,719 customers. Their comparison therefore examines how the additional behavioural variables alter the geometry presented to the clustering algorithms.

3.1.1 Data source

The empirical comparison uses the RetailRocket recommender system dataset, an event log released under the Creative Commons Attribution NonCommercial ShareAlike licence (Retailrocket, no date). The data used in the analysis contain 2,756,101 event rows, two item property files with 20,275,902 rows in total, and a category tree with 1,669 nodes. Events record an anonymous visitor, time, item and action type. Transaction identifiers are present on purchase events, but price, expenditure and revenue are absent. Event frequency is not interpreted as monetary value, and purchase rows are not described as product quantities.

The study population contains every visitor with at least one transaction event. Applying this rule to the event table yields 11,719 purchasers and 230,678 event rows for those purchasers. Both representations use these same customers in the same order. The observation snapshot is one day after the final recorded event, which gives 19 September 2015 at 02.59.47 UTC. Purchase recency is measured from this snapshot rather than from each customer's final browsing action. Stable ordering by visitor, timestamp and original row position resolves equal timestamps without making an arbitrary event sequence depend on file reading behaviour.

3.1.2 Feature engineering

Three purchase summaries form the compact representation. Purchase recency is the elapsed number of days from the last transaction event to the snapshot. Purchase occasion count is the number of distinct nonempty transaction identifiers. Purchase event count is the number of transaction rows. The latter two measures can differ when several transaction rows share an identifier. They describe purchase activity recorded in the event log and do not substitute for expenditure.

Nine additional variables expand the compact representation into the rich representation. They describe activity volume, conversion, timing and category behaviour. Table 3.1 defines all twelve variables. The three purchase variables are carried into the rich table without recalculation, so each purchaser has the same purchase summary in both representations.

Table 3.1 Recorded feature contract for the two customer representations

Feature	Block	Operational definition
Purchase recency days	Compact transaction	Days from the latest transaction event to the snapshot
Purchase occasion count	Compact transaction	Distinct nonempty transaction identifiers
Purchase event count	Compact transaction	Transaction event rows
Total events	Activity	All event rows for the visitor
Active days	Activity	UTC calendar days containing an event
Cart per view	Conversion	Add to cart events divided by view events
Transaction per view	Conversion	Transaction events divided by view events

Feature	Block	Operational definition
Activity span days	Temporal engagement	Days between first and last event
Mean interevent hours	Temporal engagement	Mean gap between adjacent events
Session count 30m	Temporal engagement	Sessions separated by more than thirty minutes
Distinct top categories	Category behaviour	Distinct root categories among known joins
Top category share	Category behaviour	Largest root category count divided by known category events

Category variables require a time-valid join because item properties change during the observation period. Each event receives the latest category property that was available at or before its timestamp. The category tree then maps that category to its root. This procedure covers 2,099,173 event rows, which is 76.16 per cent of all events. Unknown joins are excluded from category denominators rather than treated as a category. A total of 1,168 purchasers have no event with a known category. Their two category features are set to zero, while a separate coverage indicator records the missing information. This indicator is used for data checks and is not included among the twelve clustering variables. The missing category records prevent these variables from being interpreted as complete purchase preferences.

Ratio variables also use explicit denominator rules. Cart per view and transaction per view equal zero when a visitor has no recorded view event. This condition affects 428 purchasers and is retained in the feature audit. A session starts with a visitor’s first event or after an inactivity gap greater than thirty minutes. Alternative gaps of fifteen and sixty minutes are reserved for sensitivity analysis. These choices make every derived value reproducible while exposing the assumptions most likely to affect temporal structure.

Nonnegative counts and time variables receive a logarithm after adding one. Ratios and shares remain on their original scale. Each representation is then standardised feature by feature using its own fitted means and standard deviations. Full-data clustering fits these transformations on the complete cohort. Sample splitting and resampling fit them only on the relevant training or subsample rows. Principal component analysis (PCA), which replaces features with orthogonal linear combinations ordered by variance, is applied only to a separately standardised copy of the rich representation and never supplies features to rule explanations or cluster descriptions.

Before clustering, both feature tables were checked against the source data. File hashes identify the exact raw inputs. The distributed feature contract records the treatment of equal timestamps, category changes, missing categories and zero denominators; the original event files are required to independently reconstruct those joins. Each table must contain the same 11,719 unique purchaser identifiers in the same order. The three purchase variables are then compared row by row and must be identical across the two tables. These checks ensure that the compact and rich analyses differ through the nine additional behavioural variables rather than through different customers or conflicting purchase summaries.

3.2 Analytical methods and mathematical scope

The comparison uses one recorded customer cohort, feature contract, set of candidate grids, deployment constraints and evaluation measures. K Means provides a centre-based reference partition that can be approximated by ExKMC rules. CLASSIX provides a separate clustering result with native aggregation provenance. ExKMC is used here to approximate a fitted K Means reference, not as an independently selected customer clustering method; no composite score ranks it against CLASSIX. The same change in represented information is traced through each partition and its corresponding explanation.

3.2.1 Clustering candidates and selection

K Means: a partition around centres. For n observations $x_i \in \mathbb{R}^d$ ($i = 1, \ldots, n$), d denotes the number of features. K Means seeks assignments $z_i \in \{1, \ldots, K\}$ and centres $\mu_j \in \mathbb{R}^d$ ($j = 1, \ldots, K$) that minimise

$$\min_{z_1,\ldots,z_n,\, \mu_1,\ldots,\mu_K} \sum_{i=1}^{n} \|x_i - \mu_{z_i}\|_2^2.$$

The Euclidean norm $\|\cdot\|_2$ measures straight-line distance; squaring it gives more weight to large deviations. The usual alternating procedure assigns each observation to its nearest centre, then replaces each centre by the mean of its assigned observations (MacQueen, 1967; scikit-learn developers, 2025). Each step does not increase the objective, but the fitted solution need not be the global minimum. The nearest-centre regions are Voronoi cells: each contains the points closer to its centre than to any other. k-means++ initialisation favours points far from already chosen centres using squared-distance sampling; repeated initialisations reduce dependence on one starting configuration (Arthur and Vassilvitskii, 2007).

Feature scaling matters because every coordinate contributes to squared distance. Adding nine standardised behavioural dimensions changes that sum even when the original three coordinates are unchanged. Centres and their nearest-centre assignments can consequently change for the same purchasers. Standardisation removes measurement-unit differences; it does not make the choice of features inconsequential.

CLASSIX: local groups followed by merging. CLASSIX first aggregates observations into small groups and then joins groups into clusters (Chen and Güttel, 2024). The study supplies standardised observations. In the version 1.5.1 Euclidean PCA branch, the package additionally centres them and divides by the median centred Euclidean norm (using one if that scale is zero). The radius below is in these processed coordinates, not in raw feature units (Chen and Güttel, 2026).

The *principal sorting direction* is a unit vector along which the processed observations have maximum variance. An observation's *projected score* is its scalar coordinate along that direction. After sorting by these scores, the first unassigned observation becomes a *starting point*. Later unassigned observations within the *aggregation radius* of that point join its *aggregation group*. Points already assigned to an earlier group are skipped. The next unassigned point starts another group, until every observation belongs to one. A starting point is therefore an observed representative, not a fitted centroid; a group is an intermediate object, not necessarily a final cluster.

For *distance merging*, groups are linked when their starting points are within the radius multiplied by the merge scale, and connected groups share a cluster. For *density merging*, the criterion uses counts per volume in overlapping radius balls: version 1.5.1 links a pair when the intersection density is at least one of the two ball densities. It thus tests local overlap density rather than only centre separation. Minimum cluster size controls the subsequent treatment of small merged components. The recorded distance branch reassigns groups from undersized components to a nearest eligible component. In the density branch, disabling post allocation leaves rejected observations labelled -1, or *noise*. Noise denotes absence from a retained cluster under that configuration, not an invalid customer record. Small-group merging and final outlier allocation are distinct options (Chen and Güttel, 2026).

The resulting *provenance* connects each observation to its aggregation group, starting point and final label. This construction record is the reason for using CLASSIX here. It explains its own grouping process, whereas a post hoc tree approximates an already fitted reference. Neither representation is preferred in advance.

K Means uses k-means++ seeding, twenty initialisations and the fixed seed 20260801 (MacQueen, 1967; Arthur and Vassilvitskii, 2007). The number of clusters ranges from 2 to 12. CLASSIX uses Euclidean distance and sorting along the first principal direction.

Radius ranges from 0.25 to 1.75 in increments of 0.05. Minimum cluster size takes values 1, 5 and 10, while group merging uses either distance or density. The merge scale is 1.5, tiny groups can be merged, and post allocation is disabled. Each representation has 11 K Means candidates and 186 CLASSIX candidates. A recorded boundary diagnostic triggered a symmetric radius extension from 1.80 to 2.25 for both representations. The extension tests grid sensitivity and does not replace the primary selection.

Each method and representation reports an unconstrained silhouette optimum and a deployment-acceptable result. Silhouette compares an observation's mean distance to its own cluster with its smallest mean distance to another cluster; larger values indicate better separation relative to cohesion. Davies–Bouldin averages each cluster's worst ratio of within-cluster spread to centre separation, so smaller values are preferred. Calinski–Harabasz compares between-cluster with within-cluster squared dispersion, adjusted for sample and cluster counts, and larger values are preferred. These are different geometric criteria, not competing estimates of ground truth. A candidate is internally valid only when at least two nonnoise clusters remain and silhouette, Davies Bouldin and Calinski Harabasz indices are finite (Rousseeuw, 1987; Davies and Bouldin, 1979; Caliński and Harabasz, 1974). Deployment acceptability further requires 2 to 12 nonnoise clusters, no more than 20 per cent noise and no single nonnoise cluster larger than 95 per cent of nonnoise observations. Selection maximises silhouette, then prefers lower Davies Bouldin, higher Calinski Harabasz, a smaller largest cluster share, less noise, fewer clusters and finally the lexical parameter record. Runtime is measured but never selects a model. Keeping the unconstrained result reveals when an attractive internal score is produced by a partition that is unsuitable under the stated operational constraints.

Internal indices exclude observations labelled as noise, so CLASSIX index comparisons refer to each partition's nonnoise subset. Cross-representation partition comparisons retain every purchaser because a noise assignment is itself a difference in assignment. Fixed K comparisons use the same value from 2 to 12. Fixed CLASSIX comparisons use the same radius, minimum size and merge rule. Independently selected deployment partitions are compared by adjusted Rand index, arithmetic-mean normalised mutual information and variation of information (Hubert and Arabie, 1985; Meilă, 2007). ARI measures agreement in whether pairs of observations share a cluster, adjusted for the agreement expected under its fixed-size random-partition model. One means identical partitions up to labels; zero is that chance-adjusted reference, and negative values are possible. It can compare two fitted partitions without ground truth: here it measures agreement, not accuracy. Arithmetic-mean NMI divides shared label information by the mean entropy of the two partitions; VI is their sum of conditional entropies, measured in nats with natural logarithms. Higher NMI and lower VI indicate closer agreement (scikit-learn developers, 2025; Meilă, 2007). A Hungarian match finds the one-to-one relabelling that maximises overlapping assignments for transition tables (Kuhn, 1955). The resulting migration share describes changed assignments and does not imply that numeric labels identify persistent customer types.

3.2.2 Resampling and explanation evaluation

Fixed-parameter overlap-resampling stability uses ten pairs of independent 80 per cent samples without replacement. The same seed pairs are used for both representations and methods. Each sample refits its transformations and clustering with the full-data deployment parameters. Adjusted Rand index is calculated only on purchaser identifiers present in both members of a pair. The statistic measures sensitivity to sampled customers and, for K Means, initialisation, conditional on the selected configuration. It does not measure stability of model selection, calendar time or an external cohort.

The CLASSIX explanation retains public aggregation group assignments, starting point indices, final labels and cluster sizes. Merge membership is reconstructed from public group and final assignments because version 1.5.1 does not expose a documented complete edge list. Native aggregation groups use sorted rows, so the export restores input order with the saved inverse permutation before linking groups to visitors. Every exported provenance label is checked against the selected main result for the same visitor, and every representative must belong to its reported group. The trace is an explanation of the fitted CLASSIX partition and does not claim that a group boundary is a human validated customer rule.

ExKMC approximates K Means labels under five deterministic 80 per cent training and 20 per cent test splits. The primary full cohort selection gives four K Means clusters in both representations, so K is fixed at four before splitting. Log transformations, standardisation, the K Means reference and the ExKMC tree are fitted using training rows only. The test target is the label returned by the fitted K Means predictor. This estimates explanation fidelity conditional on the primary K choice rather than nested selection of K. A leaf is a terminal decision-tree region; the leaf budget caps the number of such regions, and tree depth counts tests on the longest root-to-leaf path. The leaf budget ranges from K to four times K. The main comparison fixes the budget at K, while the wider range measures the fidelity and complexity tradeoff. A supplementary analysis selects K from 2 to 12 using each training split alone. Fidelity is the proportion of exact label agreements. Complexity is reported through effective leaves, depth, path conditions and features. Rule thresholds are exported in both standardised and original units.

Technical addressability means satisfying the study's specified size, coverage and named-feature distinctiveness requirements. It is evaluated for selected CLASSIX and K Means clusters through a documented audit rather than subjective cluster names.

A nonnoise cluster must contain at least 50 customers and one per cent of nonnoise customers, retain at least 80 per cent category coverage, and differ from the global median by at least half an interquartile range on two to four named features. Passing this audit shows that a cluster can be expressed as a limited set of query conditions with adequate coverage. It does not demonstrate intervention effectiveness, revenue change or human comprehension.

3.2.3 Projection filtering and complexity

CLASSIX can rule out some candidate neighbours before computing their full Euclidean distance. The purpose of this section is to justify that computational shortcut and distinguish its theoretical cost from recorded performance. Proposition 3.1 restates the projection argument underlying CLASSIX aggregation (Chen and Güttel, 2024); it is not a new clustering objective.

Let $x_i \in \mathbb{R}^d$ be processed observation i, where d is the number of coordinates, and let u be a unit sorting direction. Define its score by $s_i = u^\mathsf{T} x_i$. A starting point is denoted by c, with score $s(c) = u^\mathsf{T} c$, and $r > 0$ is the aggregation radius in the same processed coordinates. Scores are traversed in nondecreasing order.

Proposition 3.1. *If a later point satisfies $s_i - s(c) > r$, then it cannot satisfy $\|x_i - c\|_2 \le r$.*

Proof. The Cauchy–Schwarz inequality bounds an inner product by the product of the two vector lengths. Since the projection direction has length one,

$$|s_i - s(c)| = |u^\mathsf{T}(x_i - c)| \le \|u\|_2 \|x_i - c\|_2 = \|x_i - c\|_2.$$

Thus a projected separation greater than r implies a full distance greater than r. □

In words, points already too far apart along one unit direction cannot be close in the full space. Once a sorted score exceeds the window, every later score does too, so the scan for this starting point can stop. The converse fails: a point inside the score window may be far away in other coordinates and still needs an exact distance test. Equality remains eligible in the implementation. The argument is exact in real arithmetic; it does not remove floating-point boundary effects or the order dependence of greedy aggregation.

Let n be the number of observations and q the number of resulting aggregation groups. At most $n(n-1)/2$ distinct later-point pairs can be tested during aggregation. Distance merging has at most $q(q-1)/2$ distinct starting-point pairs. Each full distance involves d coordinates, giving the conservative distance-arithmetic bound

$$O\big((n^2 + q^2)d\big).$$

This is a worst-case bound for those distance operations, not an average runtime law. Principal-direction estimation, scalar sorting and bookkeeping have additional costs; small-component reassignment can add a quadratic group-distance pass. The bound on distinct pairs need not equal the number of vectorised arithmetic operations, which may include self-comparisons.

The saved package counter counts retained aggregation distance comparisons on the tested inputs; elapsed runtime also includes preprocessing, merging and implementation overhead. Neither an observed small counter nor a short runtime proves linear complexity. The distinction matters here because the grouping trace arises during clustering itself. Its computational mechanism differs from the later construction of an ExKMC tree.

3.2.4 A local stability certificate

Section 4.4.2 changes numerical feature values slightly and measures whether CLASSIX assignments change. One possible cause of change is rotation of the principal sorting direction. This section asks a narrower mathematical question: under what sufficient conditions do the finite decisions of a distance-merging fit remain unchanged? It separates stability of a direction from stability of a complete partition.

Write S for the covariance matrix of the centred, fully processed observations, and $\lambda_1 \geq \lambda_2$ for its two largest eigenvalues. The *eigengap* $\gamma = \lambda_1 - \lambda_2 > 0$ measures how clearly the leading direction is separated from the next. A perturbation is a change of the numerical input: it gives processed observations $\widehat{x}_i$ and covariance $S + \Delta S$, where ΔS is the covariance change. Let u and $\widehat{u}$ be unit leading eigenvectors before and after this change. Choose their relative signs so that $u^{\mathsf{T}}\widehat{u} \geq 0$.

Davis–Kahan bounds the rotation of an eigenspace when its matrix changes (Davis and Kahan, 1970). Using the one-dimensional form of Yu et al. (2015, Corollary 1), and restricting attention to $2\|\Delta S\|_2 < \gamma$, gives

$$\sin \angle(\widehat{u}, u) \leq \frac{2\|\Delta S\|_2}{\gamma}.$$

Here the matrix norm is the spectral norm (the largest singular value); the angle is the acute angle between the two directions. A larger eigengap permits a smaller rotation bound for the same covariance change. If the leading directions are nearly tied, this argument gives little protection.

Let ε_x bound each processed displacement $\|\widehat{x}_i - x_i\|_2$, and let B bound every original processed norm $\|x_i\|_2$. The sign-aligned unit vectors obey $\|\widehat{u} - u\|_2 \leq \sqrt{2}\sin\angle(\widehat{u}, u)$. Consequently,

$$\begin{aligned} |\widehat{u}^{\mathsf{T}}\widehat{x}_i - u^{\mathsf{T}}x_i| &\leq \|\widehat{x}_i - x_i\|_2 + \|\widehat{u} - u\|_2\|x_i\|_2 \\ &\leq \varepsilon_x + \sqrt{2}B\frac{2\|\Delta S\|_2}{\gamma} =: b_s. \end{aligned}$$

Thus b_s is a maximum score movement, accounting for movement of both the point and the direction. The input to this bound is after all preprocessing: changes of fitted centring and scale must be included in ε_x and ΔS. A Gaussian noise standard deviation is not itself a bound on every processed displacement.

The remaining conditions protect individual decisions from crossing thresholds. Define g_s as the minimum adjacent gap in the original sorted scores. Define $g_o = |u^{\mathsf{T}}x_1|$ for the first input row used by version 1.5.1's score-sign convention. Define g_w as the smallest absolute distance of any relevant score difference from its active search-window threshold, including both aggregation radius r and distance-merge radius $1.5r$. Let g_r be the smallest absolute margin of an aggregation distance from r, and g_m the corresponding distance-merge margin from $1.5r$. These minima refer to the decisions traversed by the original fit; they must be positive for the certificate below.

Let b_r and b_m bound changes in the aggregation and merge distance tests respectively. With unchanged radii in fully processed coordinates, both may conservatively be set to $2\varepsilon_x$, by the triangle inequality. If thresholds change, their changes must also be added to the relevant bounds, including the score-window bound.

Proposition 3.2. *For a fixed distance-merging configuration and unchanged processed thresholds, the same partition is produced if*

$$g_s > 2b_s, \quad g_o > b_s, \quad g_w > 2b_s, \quad g_r > b_r, \quad g_m > b_m,$$

provided that any later nearest-eligible-group choices and tie decisions in small-component handling are also unchanged. The input identities, feature dimension, minimum-size rule and implementation branch are fixed. The claim is in exact arithmetic, or with numerical error included in the bounds.

Proof. The first condition preserves the sorted order: two scores can approach each other by at most $2b_s$. The second preserves the orientation anchor's sign, so the package's orientation agrees with the sign alignment used in the score bound. The third preserves the relevant search windows. Proceeding through the original traversal, the next unassigned starting-point identity is therefore the same; the fourth condition preserves its accept/reject decisions. Induction gives the same aggregation memberships and representative identities, although their coordinates may move. The third and fifth conditions then preserve the distance-merge links, hence their connected components and sizes. Minimum-size eligibility is consequently unchanged. The additional assumption on nearest-group choices and ties preserves any subsequent reallocation, giving the same final partition up to label names. □

Each inequality says that a decision's distance from its boundary must exceed the largest allowed movement. These requirements are deliberately conservative: failure of one does not imply a changed partition. The certificate is local, conditional and sufficient, not necessary. It is not a general proof of CLASSIX stability, and the recorded experiments do not establish that all these margins hold. Density merging additionally depends on overlap counts and volumes, which these conditions do not control. In particular, the selected rich CLASSIX fit uses density merging and is not certified by Proposition 3.2. Its empirical perturbation results in Section 4.4.2 are complementary observations, not an application of this proposition.

3.2.5 Benchmark and robustness designs

Nine labelled datasets provide an algorithm-level benchmark for CLASSIX, K Means, DBSCAN and Ward (Ester et al., 1996; Ward, 1963). Candidate selection uses only internal indices and deployment guards. Reference labels are reserved for ARI and NMI evaluation. K Means results average five fixed seeds before parameter selection. Ward, DBSCAN and CLASSIX store one structural row per candidate, while repeated fits audit label identity and runtime on the recorded environment. This empirical identity check is not a global determinism theorem.

The rich representation undergoes twelve leave-one-feature-out fits. Principal component analysis (PCA) rotates the standardised features into orthogonal directions ordered by explained variance. Here it retains at least 90 per cent of standardised variance and is compared with the named space on structure and stability, never on rule quality (Pearson, 1901; Hotelling, 1933). The CLASSIX perturbation check adds independent zero-mean Gaussian noise to each standardised feature entry, then refits with the selected parameters and compares labels with the unperturbed fit. It uses noise standard deviations 0, 0.000001, 0.0001, 0.001 and 0.01 with ten seeds at every nonzero level. Session counts are rebuilt with gaps of 15, 30 and 60 minutes. A negative control appends independently permuted rich-only dimensions to unchanged compact features. These diagnostics retain the primary selected configurations unless a section explicitly reports complete feature reselection.

Chapter 4

Results and Analysis

The results follow the evidence chain established in Chapter 3. A benchmark first checks the implementations and selection procedure. The customer analysis then examines fitted partitions, explanation and addressability before sensitivity tests probe the main observations.

4.1 Algorithm benchmark

Nine labelled datasets provide an algorithm check before the customer analysis. Six shape datasets and three multivariate datasets were evaluated with CLASSIX, K Means, DBSCAN and Ward. Selection used internal indices and deployment guards. Reference labels entered only the subsequent ARI and NMI calculation.

No method dominates the benchmark. K Means has the highest mean ARI across the nine selected results at 0.597, followed by Ward at 0.575, DBSCAN at 0.509 and CLASSIX at 0.508. CLASSIX and Ward share a mean rank of 2.44, while K Means has 2.39 and DBSCAN has 2.72. These averages conceal marked dependence on shape. CLASSIX recovers the Jain reference exactly but reaches only 0.264 ARI on R15. DBSCAN recovers Spiral exactly and performs best on Aggregation and Compound, yet its selected Wine candidate has no valid internal score. K Means is strongest on Seeds and Wine, while Ward is strongest on R15.

Observed runtime depends on implementation and dataset size. The saved timings describe the recorded machine and execution conditions; they do not establish a universal speed ordering. The benchmark instead verifies that the implementations, label-free selector and external evaluation recover varied successes and failures before the same metric code is applied to RetailRocket. The customer conclusion rests on paired evidence rather than algorithm superiority.

Table 4.1 Highest external ARI after label free selection on each benchmark dataset

Dataset	Method	ARI
Aggregation	DBSCAN	0.910
Compound	DBSCAN	0.826
Iris	CLASSIX and K Means	0.568
Jain	CLASSIX	1.000
Pathbased	CLASSIX	0.497
R15	Ward	0.804
Seeds	K Means	0.481
Spiral	DBSCAN	1.000
Wine	K Means	0.897

4.2 Representation and customer cluster structure

Changing the represented information alters the selected customer structure for both clustering methods. Table 4.2 reports the independently selected deployment-acceptable configurations. K Means selects four clusters in each representation, but the agreement between the two four-cluster partitions is only 0.014 ARI. CLASSIX selects two compact clusters and three rich nonnoise clusters with 218 additional noise assignments: these purchasers remain in the cohort but are not allocated to a retained cluster under the selected density-merging configuration. Their cross-representation agreement is 0.068 ARI. Here ARI compares two fitted assignments on the same purchasers, not either assignment against known customer types. The low values therefore indicate disagreement between representations, not low accuracy against ground truth.

Table 4.2 Deployment acceptable clustering results on the shared purchaser cohort

Representation	Method	Clusters	Noise	Largest share	Silhouette	Stability
Compact	CLASSIX	2	0	0.909	0.595	0.985
Rich	CLASSIX	3	218	0.898	0.291	0.965
Compact	K Means	4	0	0.664	0.560	0.988
Rich	K Means	4	0	0.579	0.351	0.995

The internal indices do not move in a single direction. Silhouette uses observation-level cohesion and separation; Davies–Bouldin emphasises each cluster's least favourable spread-to-centre-separation ratio; Calinski–Harabasz uses aggregate squared dispersion. Their different weighting of local separation, cluster spread and imbalance makes disagreement possible. Compact CLASSIX has silhouette 0.595, Davies Bouldin 1.057 and Calinski Harabasz 4285.5. The rich result has silhouette 0.291, Davies Bouldin 0.859 and Calinski Harabasz 1516.9. These CLASSIX values are calculated on each partition's nonnoise observations. Davies Bouldin favours the rich result, while silhouette and Calinski Harabasz favour the compact result. Compact K Means has silhouette 0.560 and

Calinski Harabasz 10459.1, compared with 0.351 and 3923.4 in the rich space. Davies Bouldin rises from 0.775 to 1.040. Richer behaviour produces a different geometry whose apparent quality depends on the criterion.

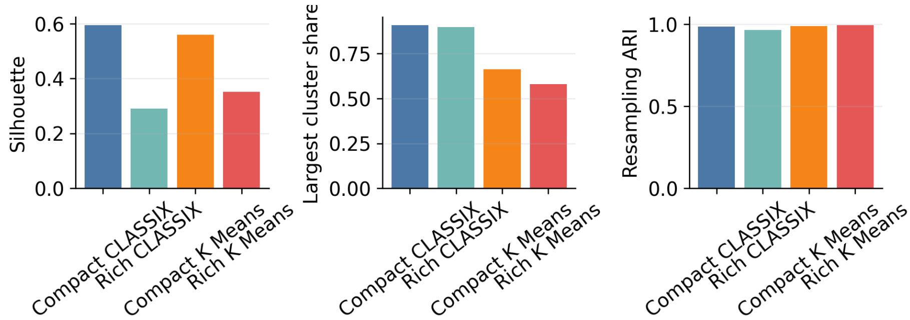

Figure 4.1 Internal structure and resampling stability for the selected customer partitions

The deployment rule materially changes the result that would be chosen by silhouette alone. Compact CLASSIX reaches silhouette 0.899 by placing 11,698 of 11,719 customers in one cluster. Rich CLASSIX reaches 0.792 with 11,716 customers in one cluster and the remaining three customers split across two tiny clusters. Compact K Means reaches 0.850 with 11,593 customers in one of two clusters. All three partitions fail the maximum share rule. Isolating a few distant observations can leave most observations much closer to their own large group than to those outliers, yielding a high silhouette without a useful subdivision of the majority. The 95 per cent cap is an operational design choice, not a mathematical law of clustering. The deployment selections sacrifice these high silhouette values to avoid reporting a nearly undivided customer base as a useful segmentation. Rich K Means is the only case where the unconstrained and deployment selections coincide.

Low agreement is not caused only by choosing different numbers of clusters. At every fixed K from 2 to 12, the K Means ARI between compact and rich representations remains between 0.012 and 0.113. At the common selected value of four it is 0.014, with NMI 0.089 and variation of information 1.691 nats. Under matched labels, 54.61 per cent of customers move to a different K Means cluster. Among 41 CLASSIX settings that are deployment acceptable in both spaces, matched-parameter ARI ranges from -0.051 to 0.392 with a median of -0.023. The independently selected CLASSIX partitions have NMI 0.054 and variation of information 0.693 nats, with 18.66 per cent assigned to a different matched cluster or noise state. The lower CLASSIX migration share partly reflects the dominance of one large cluster in both representations and should not be read as stronger substantive agreement.

Cluster sizes make the changes more concrete. Compact K Means contains 7,784, 2,494, 1,344 and 97 customers. Rich K Means contains 6,790, 3,607, 1,156 and 166. Its 166 customer cluster has a median of 338.5 total events, 28 sessions and 12 top categories, whereas the full cohort medians are 6 events, 2 sessions and 1 top category. Another rich cluster of 1,156 customers has no known category at its median and a purchase recency median of 127.4 days. These are descriptive contrasts in observed features, not customer identities or causal segments.

CLASSIX yields a sharper imbalance. The compact deployment result contains 10,656 and 1,063 customers. The rich result contains 10,331, 1,137 and 33 nonnoise customers plus 218 noise assignments. The 33-customer cluster is highly active, with medians of 869 total events, 36 sessions and 14 top categories. Its small size fails the documented addressability size rule. The 1,137-customer cluster has zero category coverage and fails the coverage rule. The dominant 10,331-customer cluster has no two sufficiently distinctive median effects. Richer features expose unusual behavioural combinations, but the selected CLASSIX partition does not turn them into a balanced set of directly addressable clusters.

Adding standardised behavioural coordinates changes pairwise distances, K Means centres and CLASSIX sorting and neighbourhood decisions. Within-space resampling changes which purchasers are fitted while retaining that representation; cross-space comparison changes the geometry itself. High repeatability and low cross-space agreement are thus compatible. This mechanism explains why such a pattern is plausible, without identifying a causal share attributable to any one feature.

Fixed-parameter overlap-resampling indicates that these structural differences are reproducible under the stated sampling scheme. Mean overlap ARI is 0.985 for compact CLASSIX and 0.965 for rich CLASSIX. The corresponding K Means values are 0.988 and 0.995. The lowest of ten pairwise values is 0.881 for compact CLASSIX and 0.941 for rich CLASSIX. Stable refitting does not imply that a partition is uniquely correct or that selection would recur. It shows that each selected representation and parameter combination tends to reproduce its own assignment pattern when one fifth of the cohort is omitted.

4.3 Explanation complexity, fidelity and addressability

The explanation results separate provenance from approximation. CLASSIX exposes how its own observations are aggregated and merged. ExKMC approximates a K Means predictor with axis aligned rules. Their numbers answer different questions and are not combined into a common explanation score.

The selected compact CLASSIX fit produces 131 aggregation groups that merge into two final clusters. The rich fit produces 559 groups and four final states consisting of three clusters and noise. Every exported customer trace contains the visitor identifier, aggregation group, representative row and final cluster. Both exports reproduce the selected full data labels exactly with ARI 1. Rich CLASSIX needs more than four times as many local aggregation groups even though its final nonnoise cluster count rises only from two to three. The larger group count means more intermediate units in the exported construction record. It describes computational provenance granularity; without a human study it cannot be translated into reading time or cognitive difficulty.

The main ExKMC comparison fixes K at four for both representations because both primary K Means selections contain four clusters. At the four-leaf budget, both trees use four effective rules with depth 3 and 2.25 conditions per rule. Compact mean training fidelity is 0.9466 and mean held-out fidelity is 0.9448. Rich fidelity is higher at 0.9639 and 0.9650. Across the five splits, compact held-out fidelity ranges from 0.9416 to 0.9480 and rich fidelity ranges from 0.9599 to 0.9697. This is descriptive split variability rather than inferential uncertainty. The close training and test values give no sign of a training-only gain that disappears on held-out customers.

Compact rules use purchase occasion count and purchase recency across all five splits. Rich rules use purchase occasion count, activity span and the number of distinct top categories. Thresholds are exported in original units after reversing the training scaler and logarithmic transform. Named thresholds support direct database style conditions, but they inherit the meaning and missingness limits of the constructed variables.

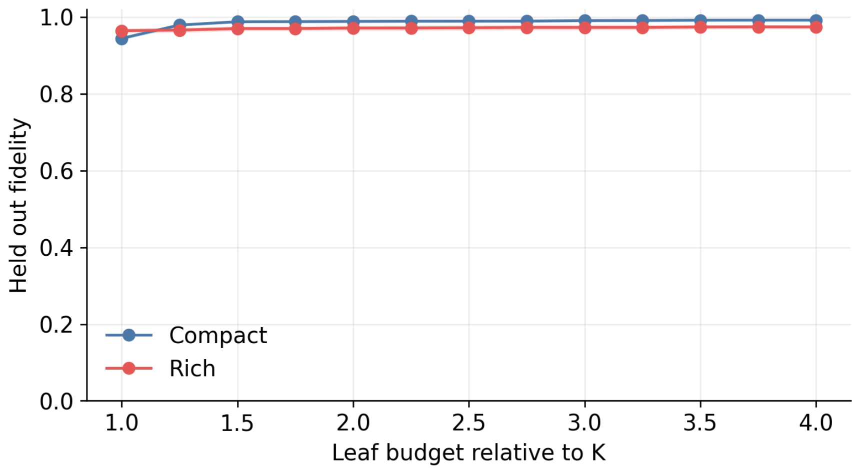

Figure 4.2 Held out ExKMC fidelity across the fixed leaf budget

Additional leaves change the two fidelity frontiers differently. At sixteen leaves, compact mean held out fidelity reaches 0.9924 with mean depth 6.6 and 4.84 conditions per rule. Rich fidelity begins higher at four leaves but reaches 0.9747 with mean depth 6.4 and 4.74 conditions at sixteen leaves. Compact fidelity begins lower and gains more from expansion, eventually exceeding rich fidelity. At the smallest budget, the fitted rich assignments are more closely approximated by the available cuts; with additional cuts, the compact reference admits a closer approximation. Representation therefore changes the observed relationship between rule budget and label agreement. This is a result for the fitted references and tree-building procedure, not a proof of minimum possible explanation complexity or human interpretability. A supplementary training only selection chooses two compact clusters and four rich clusters in every split. Its compact fidelity is 0.9985 with two rules, but that result is not used for the main complexity comparison because K differs.

The addressability audit asks a further question beyond rule export. A cluster can exist without being well separated, be repeatable without having short rules, and have short rules without adequate size or feature coverage. Passing those operational checks still does not establish business value. These properties require separate evidence. Neither compact CLASSIX cluster passes because neither differs from the cohort median on at least two named features by the required half interquartile range. In the rich result, the 33-customer cluster is too small, the 1,137-customer cluster has zero category coverage, and the 10,331-customer cluster lacks two sufficiently distinctive median conditions. None of the five nonnoise CLASSIX clusters is classified as addressable. No compact K Means cluster passes. One of four rich K Means clusters passes all gates, while the others fail because they have too many distinguishing conditions or insufficient category coverage.

Business interpretation remains narrow. Rich features expose a high-activity minority and a group with missing category histories. ExKMC shows that K Means assignments can be represented with a small number of named thresholds at high held-out fidelity. The one passing rich cluster meets a technical audit rather than an outcome test. The observed value is limited to queryability and auditable coverage. Revenue effect, campaign response and human comprehension remain unmeasured.

4.4 Sensitivity and robustness

The representation effect is not tied to one diagnostic. Its interpretation changes when the analysis preserves selected parameters, repeats selection after changing features, or destroys joint information. Fixed checks measure local dependence. Reselection repeats the candidate search and deployment rule. Permutation tests added dimensions with no aligned customer information.

4.4.1 Named features and reduced geometry

Fixed parameter deletion identifies category behaviour as the strongest local influence on the rich partitions. Removing distinct top categories reduces agreement with the full rich result to ARI 0.189 for CLASSIX and 0.516 for K Means. Removing top category share gives ARI 0.238 and 0.517. Every other CLASSIX deletion remains between 0.927 and 0.967. The corresponding K Means values remain between 0.855 and 0.991. Activity span and mean interevent time have the next largest effect on K Means, with ARI 0.874 and 0.855. Influence is unevenly distributed across the named variables, but category information alone does not generate a preferable segmentation.

The PCA diagnostic preserves 91.32 per cent of standardised rich variance in six components. It is a fixed-parameter local diagnostic rather than PCA-space model selection. K Means in this reduced space agrees with the named rich partition at ARI 0.995 and has silhouette 0.390 compared with 0.351. CLASSIX retains three clusters, lowers noise from 218 to 71 customers and agrees with the named result at ARI 0.920. Its silhouette is 0.304 compared with 0.291. Mean resampling ARI is 0.995 for reduced K Means and 0.987 for reduced CLASSIX. The principal component space preserves most fitted structure. This weakens the explanation that all twelve named dimensions must be retained for the rich partition to appear; it does not prove that redundancy has no effect or that six components are optimal. It is not used for explanation because rotated components do not retain the direct business meaning of the twelve named variables.

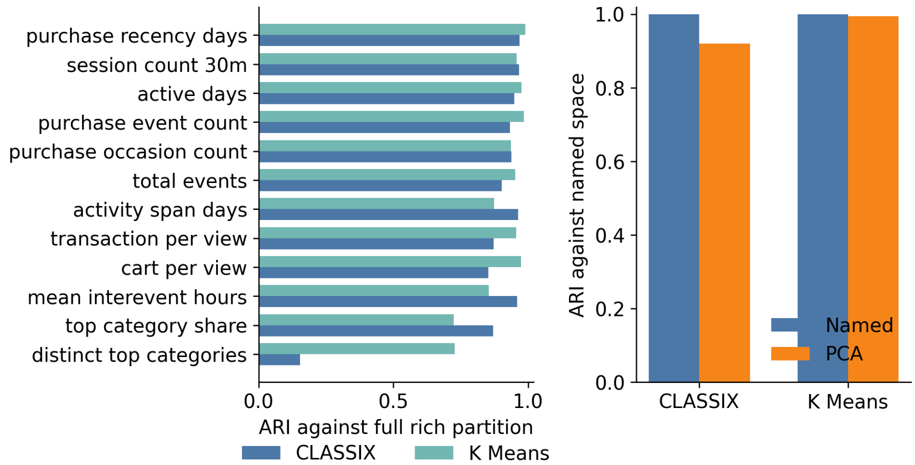

Figure 4.3 Reselected feature deletion and PCA diagnostics

Complete reselection confirms that category information creates the largest structural turn. All twenty two scenarios produce a deployment acceptable candidate. Removing distinct top categories changes CLASSIX to eleven clusters with ARI 0.153 against the full rich partition. K Means changes to two clusters with ARI 0.728. Removing the whole category block produces seven CLASSIX clusters with ARI 0.126 and two K Means clusters with ARI 0.729. The progressive endpoints reproduce the compact and rich primary partitions exactly. These 4,334 candidate fits show that the category effect persists when parameters and cluster counts are allowed to change.

CLASSIX sensitivity is distributed across the grid rather than concentrated at one radius. Deployment acceptable candidates occur at twenty three of thirty one compact radii from 0.35 to 1.45 and twenty rich radii from 0.30 to 1.25. Minimum size 1 produces no acceptable candidate in either representation. Minimum sizes 5 and 10 produce 23 and 32 acceptable compact candidates, compared with 19 and 27 rich candidates.

4.4.2 Perturbation and temporal assumptions

Small Gaussian perturbations leave the selected CLASSIX partitions almost unchanged. Compact agreement remains ARI 1 through a noise standard deviation of 0.001. At 0.01 the mean ARI is 0.999 and the minimum is 0.996. Rich agreement is also exact through 0.0001. It falls to a mean of 0.995 at 0.001 and 0.985 at 0.01, with minima of 0.985 and 0.962. The richer fit is consequently more sensitive to local numerical changes under the tested scale, but it does not become unstable in the sense of wholesale reassignment. These finite perturbation trials show observed numerical robustness at the tested scales, not a certificate for all nearby inputs. The rich selection uses density merging. The mathematical certificate applies only to distance merging when its stated margins hold.

Reconstructing sessions from raw events gives a similar qualified result. The selected thirty minute window reproduces both partitions exactly. A fifteen minute window gives ARI 0.976 for K Means and 0.971 for CLASSIX. A sixty minute window gives ARI 0.985 and 0.959. The four cluster K Means result and the three cluster CLASSIX result persist at both alternative windows, although the CLASSIX noise count changes from 218 to 220 and 239. The conclusions do not arise solely from choosing exactly thirty minutes, while the detailed assignments still depend on the session convention.

4.4.3 Dimensionality control

The permutation control sets a stricter boundary on causal interpretation. The compact columns remain unchanged while the nine additional dimensions are independently permuted across customers. These columns preserve their marginal distributions but destroy their joint relationship with the compact representation and with each other. With the compact K Means setting held fixed, adding all nine permuted columns lowers agreement with the compact partition to ARI 0.017 and silhouette to 0.127. With the compact CLASSIX setting held fixed, adding even one permuted dimension produces a single cluster and ARI 0.

The single deterministic permutation control is an existence demonstration rather than a variance decomposition. Standardisation gives each added variable comparable weight, so random added dimensions can alter distances, principal directions and radius neighbourhoods in at least one realised order. The control does not estimate how much of the observed compact-to-rich difference is caused by dimensionality. Named feature deletion and PCA separately show that category and temporal information contribute reproducible structure in the observed rich representation.

4.5 Integrated discussion and validity limits

Strongly supported findings The first research question is answered by the paired assignments: representation materially changes the observed partitions for both K Means and CLASSIX on the same purchasers. Matched-parameter comparisons show that different selected cluster counts alone do not account for the disagreement. High fixed-parameter resampling agreement within each space is compatible with this result because it repeats a fit within one geometry. It does not test whether two geometries should produce the same groups.

The second question has distinct answers for the two explanation objects. The rich CLASSIX fit has a more detailed aggregation record. ExKMC's relative fidelity changes with the leaf budget, so richer features do not uniformly simplify the approximation. The single passing rich K Means cluster also shows why assignment, geometric separation, repeatability, short rules and operational addressability must be evaluated separately. None of these findings establishes commercial benefit.

Conditionally supported findings The third question concerns sensitivity within the recorded design. Deletion and reselection identify strong dependence on category features; PCA preserves much of the fitted structure after reducing dimension. Small Gaussian changes and alternative session windows retain substantial agreement with the original rich fit. Each diagnostic tests a specified change, not every plausible preprocessing choice. The permuted-feature control demonstrates that added dimensions can alter grouping without aligned customer information; it cannot allocate the observed effect between dimension and meaningful behaviour.

The deployment constraints prevent nearly single-cluster solutions from dominating selection, but their thresholds reflect this study's operational choices. Likewise, fixed-parameter overlap-resampling measures conditional repeatability, not the stability of the complete selection procedure. The local mathematical certificate supplies sufficient distance-branch conditions; no claim is made that the saved fits satisfy all its margins, and it cannot certify the rich density-branch result. The labelled benchmarks provide external checks on several shapes and dimensions, not a guarantee of transfer to purchaser behaviour.

Unresolved questions and evidence limits The study concerns purchasers in one anonymous event log and excludes visitors without a transaction event. It contains no demographic, price, revenue, campaign or response data. Category joins cover 76.16 per cent of events, while 1,168 purchasers have no known category. Zero ratios can reflect absent recorded views rather than a substantive conversion pattern. These limits prevent labels such as valuable, loyal or profitable from being inferred from the clusters.

CLASSIX provenance records its own fitted grouping; ExKMC fidelity measures agreement with a K Means predictor. Neither verifies that predictor against objectively correct customer types or measures human comprehension. Addressability checks size, coverage and named-feature distinctiveness. Prospective campaign outcomes and user evaluation would be needed to assess realised value and understanding, but they are outside the research questions answered by these data. The supported contribution is the separation and joint examination of these properties under a change of representation, not the universal superiority of richer features.

Chapter 5

Conclusion

For the same 11,719 RetailRocket purchasers, compact and rich features produce substantially different K Means and CLASSIX assignments. Matched-parameter comparisons also show disagreement, so the result does not arise only from selecting different cluster counts. High resampling agreement shows that each configuration repeats its own grouping, without implying agreement across feature spaces.

Representation also changes the explanation evidence. Rich ExKMC fidelity begins higher at four leaves, but compact fidelity exceeds it at sixteen leaves. Thus the relative accuracy of these rule approximations depends on the permitted complexity. The rich CLASSIX fit has more intermediate aggregation groups, which describes the detail of its construction record rather than measured human difficulty. One rich K Means cluster passes the size, coverage and distinctiveness audit; no compact K Means or CLASSIX cluster passes. Accurate rule approximation and repeatable clustering therefore do not by themselves establish operational addressability.

The sensitivity analyses qualify these findings under feature deletion, reduced dimension, numerical perturbation and alternative session windows. They do not show that every representation choice would preserve the same result. The projection argument explains a valid computational shortcut, while the local stability proposition gives sufficient conditions only for a fixed distance-merging fit. It does not certify the selected rich density-merging result.

Representation is part of the empirical clustering model rather than a neutral preprocessing choice. In this dataset, richer behavioural information changes geometry, assignments and explanation properties, without consistently improving clustering quality, simplifying explanations or producing deployment-ready groups. The study does not establish objectively correct customer types, human comprehension or realised business value. Those questions require evidence beyond the recorded event log.

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Appendix A

Technical Reproduction Map

The reported experimental results are reproduced from the distributed data with `python-msrc.reproduce`. The fixed sequence covers the benchmark, customer clustering, explanations, sensitivity analyses, summary tables and figures. The optional `--compile-pdf` flag also rebuilds the report after rerunning the experimental sequence. To compile only the manuscript while retaining saved results, run `xelatex-interaction=nonstopmode-halt-on-error-output-directory=buildmain.tex` three times from `dissertation/overleaf`. Environment versions and random seeds are recorded in `environment.yml` and `protocol/analysis_protocol.json`.

Primary clustering selections are stored in `results/retailrocket/selected_configurations.csv`. Customer level labels, transitions, profiles and resampling records remain separate so that every reported aggregate can be recalculated. ExKMC training selection, held out fidelity and exported rules are stored in distinct files. CLASSIX provenance includes an exact label audit against the selected result.

The sensitivity directory separates fixed parameter diagnostics from complete candidate reselection. PCA records that reduced components are not permitted for rules. The zero perturbation rows have ARI 1. Session windows are reconstructed from the distributed purchaser timestamp table. Permuted dimensions retain the compact columns exactly while destroying their joint customer alignment.

The distributed feature tables and purchaser event timestamps are sufficient for the reported experiments. The optional `--rebuild-features` flag reconstructs these tables when the four original RetailRocket files are available. The repository `README.md` records the execution sequence, output map and licence boundary.